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Supporting Robotic Process Automation through Natural Language Processing

Received August 31, 2020

Abstract: Natural Language Processing (NLP) is a field concerned with the analysis of natural language from a computational perspective. NLP techniques cover a wide range of tasks, such as information extraction, classification, and semantic analysis. As such, NLP has considerable potential to support Robotic Process Automation (RPA) efforts. In particular, we use this chapter to highlight selected opportunities where a successful application of NLP techniques can support various stages of the RPA lifecycle. Therefore, we consider the potential of NLP for the identification of RPA opportunities, the design of RPA routines, as well as their actual execution. We discuss how these opportunities can be supported through the application of existing techniques and, furthermore, highlight open research challenges.

Keywords: Robotic process automation, natural language processing

1 Introduction

Many organizations currently face the challenge of keeping up with increasing digitization. Among others, it requires them to adapt existing business models and to respectively improve the automation of their business processes [32]. While the former is a rather strategic task, the latter calls for specific operational solutions. One of the most recent developments to increase the level of automation is referred to as Robotic Process Automation (RPA). In essence, RPA emerges in the form of software-based solutions that automatically execute repetitive and routine tasks [8]. In this way, knowledge workers can dedicate their time and effort to more complex and value adding tasks.

Natural Language Processing (NLP) is a field concerned with the analysis of natural language from a computational perspective [21]. NLP techniques cover a

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wide range of tasks, such as information extraction, classification, and semantic analysis. In the last decade, the maturity achieved by NLP technology, together with the explosion of big data and deep learning techniques, have turned the spotlight to the possibilities offered by NLP approaches for a variety of novel applications. Specifically, the potential of NLP to synthesize information from unstructured and semi-structured sources, such as textual documentation and free-text comments, makes it a prime candidate to support tedious, time-consuming tasks in organizational settings [1].

As such, we believe that NLP has a lot to offer in the context of RPA. Therefore, we use this chapter to outline various opportunities that NLP provides for different tasks in an RPA lifecycle. In particular, we discuss the application of NLP for:

1. *Identifying RPA opportunities.* NLP techniques can be leveraged to identify potential automation candidates through, e.g., semantic analysis of process documentation and the identification of re-occurring patterns.
2. *Designing RPA routines.* To implement RPA routines, tasks in an organizational process need to be transformed into a formalized, automatically executable representation format. NLP techniques can support this endeavor through, e.g., the automated extraction of formal specifications from textual resources, such as work instructions.
3. *Executing RPA routines.* Finally, NLP techniques can be used during the actual execution of implemented RPA routines. In this manner, RPA can become *smarter* and go beyond the automation of simple tasks [6]. For instance, by automating process steps involving the actual interpretation of unstructured input, such as e-mails from clients.

The remainder of this chapter is structured as follows. Section 2 discusses how NLP techniques are already applied in the closely related area of Business Process Management (BPM). Then, Section 3 shows how these and other NLP-based techniques can be lifted to the context of RPA. Finally, Section 4 concludes the chapter and highlights open research directions.

2 Background

In this section, we reflect on NLP-based approaches developed to target several use cases in the context of Business Process Management (BPM). BPM is a research area concerned with the analysis of how work is performed in an organization. The overall aim is to ensure consistent process outcomes and to take advantage of improvement opportunities [15]. As such, there is a clear relation between the goals and methods of BPM and RPA [6].

In this section, we discuss various use cases of applying NLP to BPM, which can be lifted to RPA settings as well. We divide this discussion into the analysis of text inside of process models (Section 2.1) and the analysis of text outside of process models, i.e., in supplementary resources (Section 2.2).

2.1 Analyzing Text Within Process Models

In this section, we first introduce the notion of process models. Then, we illustrate the analysis of text within process models through label parsing and the use cases that parsing opens up.

Process models. Process models are the de facto standard to capture organizational processes in a structured manner. As such, they form the central artifact for the vast majority of process analysis techniques, including those described in this chapter.

Various notations are used to define process models, of which Petri nets represent the most common formal model, whereas EPCs (Event-driven Process Chains) [36] and BPMN (Business Process Model and Notation) [13] represent the most commonly adopted methods in practice. Figure 1 depicts an exemplary BPMN model. The rounded rectangles denote the four *activities* in the process, whereas the circles are used to, respectively, depict the start and end *event* of the process. Finally, the diamond shapes with a plus-sign indicate that the two activities in between them can be executed *concurrently*.

The gray nodes are semantic annotations of the activity labels in the model. They are not part of BPMN models but result from the NLP-based analysis technique discussed next.

Label parsing. The parsing of process model activity labels focuses on the recognition of several semantic components in the labels. In particular, parsers generally aim to extract the *action*, e.g., “*obtain*”, the *business object* to which this action relates, e.g., “*process data*”, and some *additional* information fragment, e.g., “*from the ERP system*”. Such semantic annotations are depicted for the activities in Figure 1. Automatically obtaining these annotations is a challenging task, however. The two activities “*Process performance analysis*” and “*Sending performance report to inquirer*” illustrate this vividly. Both activity labels are short, do not represent proper sentences (they do not contain a verb), and the underlying syntactical patterns differ. In the former case, the action is provided as a noun at the end of the label (“*analysis*”); in the latter case, the action is provided as a noun in the beginning (“*sending*”). Because of these factors, standard NLP tools, such as the Stanford Parser, are not applicable [24]. However, there are dedicated

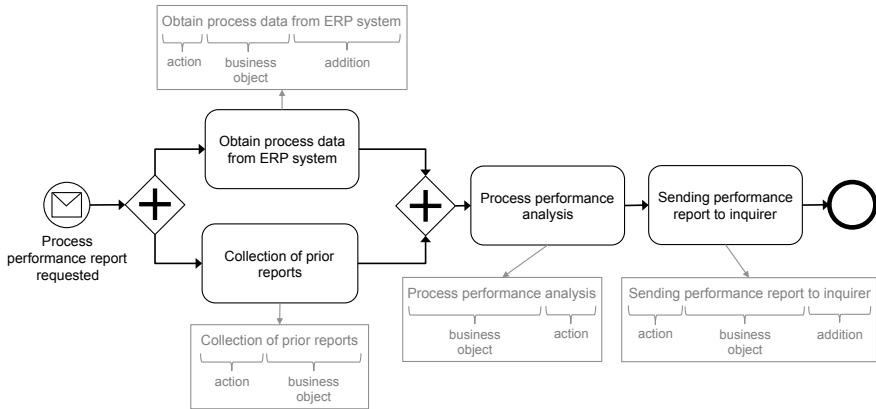


Fig. 1: BPMN model example with annotated activity labels (from [25])

label annotation techniques available. Among others, there is a heuristics-based technique exploiting the label context [31] and a technique building on Hidden Markov models [25].

Label-based analysis. The availability of activity label annotations opens the door for several advanced analyses. For example, the annotations allow to automatically check different quality criteria of process models, such as naming conventions [28] and linguistic consistency [34]. They also provide the basis for the identification of recurring patterns in process model collections [30] and for the detection of commonalities and differences between process models [40].

2.2 Analyzing Process-Related Information in Textual Documents

On top of the analysis of text within process models, a range of existing works targets the analysis of process-related information contained in different kinds of textual resources. A majority of these focus on the extraction of process models from such textual documents, whereas others target situations in which textual documents exist alongside of process models themselves.

Extracting process models from texts. Given their potential as a basis for various analysis techniques, the benefits of documenting organizational processes using process models are clear. However, in practice, process information is often spread over informational artifacts beyond these models [5]. Realizing that some stakeholders struggle to read and interpret process models, organizations have

recognized the value of maintaining text-based process descriptions in place of or alongside model-based ones [29]. While such textual descriptions may not be suitable to represent complex aspects of a process in a precise manner, they can be created, maintained, and understood by virtually everyone.

In order to make the information contained in textual documentation suitable for process analysis techniques, various approaches have been developed that elicit process models from different kinds of texts, so-called *text-to-model* transformation approaches. Some approaches target specific kinds of documents, such as the generation of models from use cases [38], group stories [19], or methodological descriptions [16], while others take general imperative [17, 18] and declarative [2] process descriptions as input. These techniques combine extensive use of general-purpose NLP tools with specifically tailored analysis techniques in order to extract process-related information from natural language texts. Nevertheless, due to the complexity of the overall challenge and the inherent ambiguity of natural language [4], the process models obtained in this manner typically require manual verification [37].

Analyzing supplementary materials. Recognizing that textual documents often exist alongside of process models, various techniques have been developed that target both model and text at the same time. For instance, different approaches [3, 35] have been established that align and compare process models against textual descriptions in order to ensure consistency across representations. Other works exploit the additional execution details provided by *work instructions* that are associated with process models. This information is, for instance, employed to improve the alignment of process models to recorded process executions [10] and to increase the recall in process querying [26].

3 Scenarios for NLP in RPA

In this section, we will illustrate how NLP can contribute to the identification of RPA opportunities (Section 3.1), the design of RPA routines (Section 3.2), and the execution of routines themselves (Section 3.3).

3.1 NLP for RPA Opportunity Identification

With RPA opportunity identification we refer to the identification of routines (either processes or parts thereof) that are good candidates to be automated through RPA bots. Existing RPA tools provide limited support for this challenge. As a

result, this task often requires a considerable amount of manual effort, by means of interviews, observation, and analysis of documentation [7].

Several recent research directions aim to support this task through the analysis of event logs. For instance, Jimenez-Ramirez et al. [20] propose a method that incorporates process mining techniques into the early stages (including opportunity identification) of the RPA lifecycle, whereas Bosco et al. [11] present a concrete method to detect deterministic routines in user interaction logs. Complimentary to such approaches, NLP techniques can be used in various manners to support the identification of RPA opportunities. Specifically, existing work can primarily support opportunity identification through the automatic detection of user tasks and the recognition of recurring process patterns:

Automatically detecting automation candidates. Prime candidates for RPA are so-called *user tasks*, which are defined as process steps in which a user interacts with an information system [15]. Such user tasks are likely to be good automation candidates because the fact that a user interacts with an information system makes it more likely that the associated manual effort can actually be emulated by an RPA bot.

In earlier work, we developed an approach [27] to automatically identify these user tasks in process repositories. This approach employs NLP techniques to first extract activities (i.e., process steps) from various available resources, such as process models and textual work instructions. Afterwards, the approach classifies the activities according to their degree of automation, i.e., recognizing whether an activity is manual, a user task, or automated. While not all user tasks identified in this manner may necessarily be suitable for automation, the approach represents a valuable filtering technique that can serve as a starting point for the identification of opportunities, even in the presence of vast process repositories.

Similarly, we have developed an approach that allows users to query process repositories, covering both process models and textual documents, for specific behavioral patterns. In the context of RPA opportunity identification, this can be employed to easily retrieve all cases in which a specific activity routine occurs. For instance, users can query all cases in which a user fills in an electronic form based on information retrieved from a customer order, which is a typical target for automation efforts. While this requires some input based on domain knowledge as opposed to the previous fully automated classification approach, it has the benefit that the obtained results are more specific and, therefore, more accurate.

A primary benefit shared by these two approaches is that their incorporation of NLP techniques allows them to identify RPA opportunities in textual documentation associated with (parts of) processes. As shown in [26], the number of relevant results that can be obtained by also querying these documents can increase considerably.

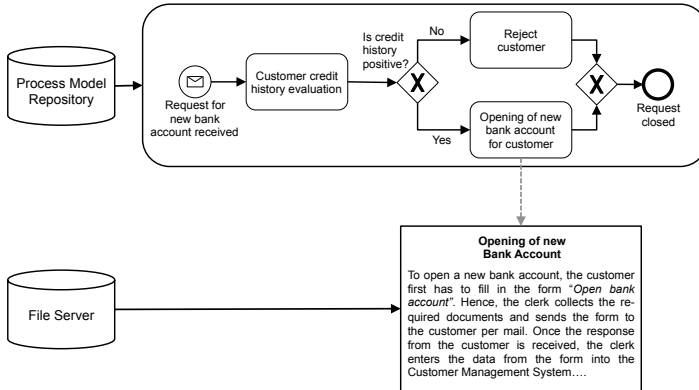


Fig. 2: Process model and associated work instruction (from [26])

That is, because process models typically provide a relatively high level view on a process. By contrast, information relevant to the identification of RPA opportunities, such as how exactly a process step is currently performed, is typically stored in more detailed work instructions. For instance, in Figure 2, only by considering the work instruction associated with the “*Opening of a new bank account for customer*” activity, it becomes clear that this activity involves the entering of information into a form and forwarding it in a mail. By analyzing this instruction using NLP techniques, it would thus be possible to recognize that this process step provides a promising candidate for automation. This opportunity would be missed without employing natural language analysis.

Detecting recurring patterns. In organizations with dozens or hundreds of processes, efforts to automate process steps using RPA are, arguably, best invested into commonly occurring routines. That is, because the return on investment tends to be bigger when automating routines that recur in various processes throughout an organization.

Such recurring patterns can be recognized by detecting commonly occurring behavioral patterns in process repositories. Without employing NLP techniques, one could detect these when there are fully identical fragments present in multiple process models. To illustrate this, consider the process model fragment in Figure 3a, which depicts a part of a process model concerned with receiving and responding to meeting requests. It is well-imaginable that this procedure is relevant to a variety of processes in an organization. For example, it may be part of any process in which a third party, such as a customer, supplier, or another department, needs to schedule a meeting. If this exact fragment is used in all these processes, it would be straightforward to recognize this recurring pattern.

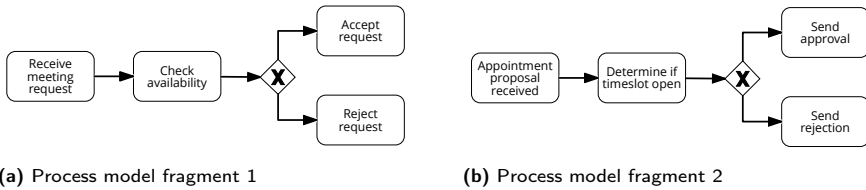


Fig. 3: Semantically similar process model fragments

However, the potential of such pattern detection can be greatly improved by employing NLP techniques. Consider the model fragment depicted in Figure 3b, which describes an equivalent process fragment, though with very different terminology from Figure 3a. By employing NLP techniques, one would be able to automatically recognize that these fragments are closely related, for instance, by recognizing the *semantic similarity* between terms such as “meeting” and “appointment” and “accept” and “approval”. So-called *process model matchers* [40] address this task by considering a variety of factors that may indicate similarity between parts of processes, including syntatic similarity, synonyms, and hypernyms. In this manner, NLP techniques make the detection of re-occurring patterns much more robust to terminological differences that are commonly part of real-world process model collections. By employing label parsing techniques, as discussed in Section 2.1, pattern detection may even be used to recognize re-occurring patterns involving different business objects [39].

Overall, the two general use cases discussed in this section should be regarded as complementary. The former use case targets the detection of specific process model activities or fragments that are relatively easy to automate, whereas the latter use case focuses on the detection of re-occurring patterns that make the investment required for automation more worthwhile.

3.2 NLP for RPA Routine Design

Once a suitable opportunity for automation has been identified, the next step is to design the RPA routine that can emulate the currently manual actions. This involves the formalization of the procedure, which typically is achieved by a human designer who establishes a flowchart. This flowchart captures the actions to be performed by an RPA bot on a specific system [7].

NLP techniques, in particular the text-to-model transformation techniques discussed in Section 2.2, can provide valuable support to human designers in this regard. The transformation techniques can be applied to textual documents

capturing an entire process, but may also be applied on work instructions, which provide detailed explanations of how specific steps of a process should be executed. The latter is particularly promising, since these work instructions are much more likely to provide the details necessary for the automation of a certain step than process models [5].

The choice for a particular technique to be employed for this purpose depends on the characteristics of the textual document(s) used as input. Documents that provide a chronological description of a process (or process step), can best be handled by techniques that extract imperative process models from text, such as the state-of-the-art technique by Friedrich et al. [17]. However, for documents that are closer to a requirements or regulatory specification, techniques to extract declarative process models [2] or business rules [41] may be more suitable.

In any case, the process models extracted by these NLP-based techniques should be regarded as a helpful starting point for the formal definition of automation routines. On the one hand, manual analysis is required to ensure that the obtained models are correct and to resolve any ambiguity [4, 37]. On the other hand, this step only covers the extraction of the as-is state of the process, including the steps that are currently performed manually. How to automate the interaction with an information system will most likely still require manual input or a combination with other techniques.

3.3 NLP in Routine Execution

Finally, we consider the actual execution of RPA routines. The value of NLP to enhance this task is already widely recognized and part of what both scholars and tool vendors refer to as *cognitive RPA* [22]. While cognitive RPA is not limited to the application of NLP in routine execution, NLP plays a major role due to the general prevalence of textual content in organizational processes. This is also reflected by specialized RPA tool vendors, such as Lexalytics¹, Expert System², or RoboRana³, which do not only use but also advertize building on NLP techniques.

The use cases for NLP in the context of routine execution can be divided into two main categories. The first relates to the automated analysis of *semi-structured content*, which can be, among others, found in invoices, insurance claims, and contracts. While such documents contain free natural language text, the documents adhere to an overall structure and typically have a number of specific data items

¹ <https://www.lexalytics.com/>

² <https://expertsystem.com/>

³ <https://roborana.be/en/>

(e.g., the final amount on an invoice) that provide a certain degree of structure and orientation. The second category relates to analysis of *unstructured content*, which can be typically encountered in customer e-mails, phone calls, chats, and social media posts. These sources need to be expected to be free of any regular structure. In the following, we discuss three major NLP use cases in more detail: the first relating to the the analysis of semi-structured content and the latter two to the analysis of unstructured content.

Invoice processing. A relevant use case relating to the analysis of *semi-structured content* is the automated processing of invoices. Based on semi-structured document parsers, a routine could provide the possibility to automatically 1) extract relevant data (such as the invoicing company, products, services, amounts, and dates), 2) store the extracted data in a database or a file, and 3) trigger an automated payment. In case an invoice is not available as a document that can be parsed (e.g., consider PDFs created from scans), Optical Character Recognition (OCR) techniques can be used in the pre-processing phase. It is interesting to point out that the technology required for extracting data from semi-structured text documents is not new. Many techniques and patents date back to the early and mid 2000s (see e.g. [12, 23]). However, the combination with existing RPA capabilities provides a new context for their use.

Ticket classification. A frequently discussed use case relating to the analysis of *unstructured content* is the automated classification of tickets. Since many organizations face hundreds or even thousands of support tickets every day, automated support can be expected to come with considerable time savings. Such automated support may be as simple as using document classification techniques to determine what a ticket is about and what problems the person is having. As a result, organizations can benefit from faster triage and faster allocation to the right expert. While document classification techniques have been around for some time, particularly the automated classification of tickets is a rather recent an active area of research [9, 33].

Automated customer service. A use case that is taking ticket classification a step further is automated customer service. The main observation behind this use case is that many customer requests relate to rather trivial matters. If it is possible to set up an RPA routine for some of these trivial matters (e.g., for changing an address or changing the seat on an aircraft), it would be possible to build a system that either a) matches a request to an existing routine or b) decides that human intervention is required. From a technological perspective, such a system is challenging because it needs to interact with the customer automatically (e.g., to obtain further information or validate the customer's ID). An example for such

a system is the chat bot system of the Dutch airline KLM called *DigitalGenius*⁴. DigitalGenius was set up to deal with customer inquiries that would be typically found in tickets. According to KLM, DigitalGenius supports up to 50% of all customer inquiries, leading to a significant reduction in both the human effort required and response time for the client.

The three use cases described above illustrate that existing NLP techniques provide interesting opportunities to enhance traditional RPA capabilities. The key challenge for the successful application in practice is to properly address the problem of accuracy: NLP-based techniques simply cannot be expected to always deliver perfect results. This means that NLP-based RPA routines need to be able to decide when they are unable to accurately process a certain task and need to include a human in the loop.

4 Conclusion

In this chapter, we highlighted the potential of employing NLP techniques to support various tasks in the RPA lifecycle. In particular, we first showed how NLP techniques can help to identify suitable candidates for automation through the identification of so-called user tasks and by identifying re-occurring process patterns. Second, we illustrated how text-to-model transformation approaches can be used as a first step in the formalization of RPA routines by automatically extracting structured process information from textual documentation. Finally, we considered how NLP can contribute to the actual execution of RPA routines themselves. We discussed use cases that involve the automation of routines dealing with both semi-structured and unstructured content. Here, we specifically illustrated how NLP techniques can enable the automation of tasks related to invoice processing, ticket classification, and automated customer service.

Maturity. The maturity of state-of-the-art approaches for all three tasks leaves considerable opportunities for further improvement:

- The use of NLP for the identification of RPA opportunities is still relatively new. Out of the various techniques developed to support the identification of RPA opportunities, only the few approaches discussed in Section 3.1 incorporate considerations of natural language. As such, there are still substantial opportunities for further work in this direction, e.g., through the development of new NLP-driven identification approaches or by combining NLP-driven approaches

⁴ <https://www.digitalgenius.com/>

with approaches targeting other process perspectives, such as control-flow and data transformations.

- The use of NLP to support routine design, as discussed in Section 3.2, is driven by existing work on the transformation of textual descriptions into formal process specifications. These current approaches are primarily heuristics-based, which results in problems when dealing with highly flexible textual documents from practice. However, recent advances in NLP, such as deep bidirectional transformers [14], provide a promising endeavor for future development in order to move away from heuristics-based transformation and to, therefore, overcome limitations of the current state-of-the-art.
- Finally, NLP has been applied to support the execution of RPA routines in various manners, though the potential for further development is considerable. Such development can focus on both the design of more advanced techniques to provide better solutions in existing scenarios, as well as focus on the usage of NLP to enable novel scenarios in which RPA can be employed.

Outlook. Although the benefits of applying NLP in all three discussed tasks are apparent, we believe that the main potential for future work lies in the application of NLP to routine execution itself. While the application of NLP for the first two tasks, opportunity identification and routine formalization, reduces the manual effort required for these tasks, the application of NLP for routine execution can truly expand the task's potential and, thus, broaden the potential of RPA as a whole. By employing NLP techniques for this purpose, RPA can thus be transformed from a rudimentary to an intelligent automation tool.

References

- [1] van der Aa, H., Carmona Vargas, J., Leopold, H., Mendling, J., Padró, L.: Challenges and opportunities of applying natural language processing in business process management. In: COLING, pp. 2791–2801. Association for Computational Linguistics (2018)
- [2] van der Aa, H., Di Ciccio, C., Leopold, H., Reijers, H.A.: Extracting declarative process models from natural language. In: International Conference on Advanced Information Systems Engineering, pp. 365–382. Springer (2019)
- [3] van der Aa, H., Leopold, H., Reijers, H.A.: Comparing textual descriptions to process models—the automatic detection of inconsistencies. *Information Systems* **64**, 447–460 (2017)
- [4] van der Aa, H., Leopold, H., Reijers, H.A.: Checking process compliance

- against natural language specifications using behavioral spaces. *Information Systems* **78**, 83–95 (2018)
- [5] van der Aa, H., Leopold, H., van de Weerd, I., Reijers, H.A.: Causes and consequences of fragmented process information: Insights from a case study. In: AMCIS (2017)
- [6] van der Aalst, W.M., Bichler, M., Heinzl, A.: Robotic process automation (2018)
- [7] Agostinelli, S., Marrella, A., Mecella, M.: Towards intelligent robotic process automation for bpmers. arXiv preprint arXiv:2001.00804 (2020)
- [8] Aguirre, S., Rodriguez, A.: Automation of a business process using robotic process automation (RPA): A case study. In: Workshop on Engineering Applications, pp. 65–71. Springer (2017)
- [9] Altintas, M., Tantug, A.C.: Machine learning based ticket classification in issue tracking systems. In: Proceedings of International Conference on Artificial Intelligence and Computer Science (AICS 2014) (2014)
- [10] Baier, T., Mendling, J., Weske, M.: Bridging abstraction layers in process mining. *Information Systems* **46**, 123–139 (2014)
- [11] Bosco, A., Augusto, A., Dumas, M., La Rosa, M., Fortino, G.: Discovering automatable routines from user interaction logs. In: International Conference on Business Process Management, pp. 144–162. Springer (2019)
- [12] Chieu, H.L., Ng, H.T.: A maximum entropy approach to information extraction from semi-structured and free text. *Aaai/iaai* **2002**, 786–791 (2002)
- [13] Chinosi, M., Trombetta, A.: Bpmn: An introduction to the standard. *Computer Standards & Interfaces* **34**(1), 124–134 (2012)
- [14] Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: BERT: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805 (2018)
- [15] Dumas, M., La Rosa, M., Mendling, J., Reijers, H.A., et al.: Fundamentals of business process management, vol. 1. Springer (2013)
- [16] Epure, E.V., Martín-Rodilla, P., Hug, C., Deneckère, R., Salinesi, C.: Automatic process model discovery from textual methodologies. In: Research Challenges in Information Science (RCIS), 2015 IEEE 9th International Conference on, pp. 19–30. IEEE (2015)
- [17] Friedrich, F., Mendling, J., Puhlmann, F.: Process model generation from natural language text. In: International Conference on Advanced Information Systems Engineering, pp. 482–496. Springer (2011)
- [18] Ghose, A., Koliadis, G., Chueng, A.: Process discovery from model and text artefacts. In: Services, 2007 IEEE Congress on, pp. 167–174. IEEE (2007)
- [19] de Gonçalves, J.C., Santoro, F.M., Baiao, F.A.: Business process mining from

- group stories. In: Computer Supported Cooperative Work in Design, 2009. CSCWD 2009. 13th International Conference on, pp. 161–166. IEEE (2009)
- [20] Jimenez-Ramirez, A., Reijers, H.A., Barba, I., Del Valle, C.: A method to improve the early stages of the robotic process automation lifecycle. In: International Conference on Advanced Information Systems Engineering, pp. 446–461. Springer (2019)
- [21] Jurafsky, D.: Speech & language processing. Pearson Education India (2000)
- [22] Lacity, M., Willcocks, L.P.: Robotic process and cognitive automation: the next phase. SB Publishing (2018)
- [23] Lemon, M.J., Castellanos, M., Stinger, J.R.: Method for content mining of semi-structured documents (2005). US Patent 6,912,555
- [24] Leopold, H.: Natural language in business process models. Ph.D. thesis, Springer (2013)
- [25] Leopold, H., van der Aa, H., Offenbergh, J., Reijers, H.A.: Using hidden markov models for the accurate linguistic analysis of process model activity labels. *Information Systems* **83**, 30–39 (2019)
- [26] Leopold, H., van der Aa, H., Pittke, F., Raffel, M., Mendling, J., Reijers, H.A.: Searching textual and model-based process descriptions based on a unified data format. *Software & Systems Modeling* **18**(2), 1179–1194 (2019)
- [27] Leopold, H., van der Aa, H., Reijers, H.A.: Identifying candidate tasks for robotic process automation in textual process descriptions. In: Enterprise, business-process and information systems modeling, pp. 67–81. Springer (2018)
- [28] Leopold, H., Eid-Sabbagh, R.H., Mendling, J., Azevedo, L.G., Baião, F.A.: Detection of naming convention violations in process models for different languages. *Decision Support Systems* **56**, 310–325 (2013)
- [29] Leopold, H., Mendling, J., Polyvyanyy, A.: Supporting process model validation through natural language generation. *IEEE Transactions on Software Engineering* **40**(8), 818–840 (2014)
- [30] Leopold, H., Pittke, F., Mendling, J.: Automatic service derivation from business process model repositories via semantic technology. *Journal of Systems and Software* **108**, 134–147 (2015)
- [31] Leopold, H., Smirnov, S., Mendling, J.: On the refactoring of activity labels in business process models. *Information Systems* **37**(5), 443–459 (2012)
- [32] Leyh, C., Bley, K., Seek, S.: Elicitation of processes in business process management in the era of digitization—the same techniques as decades ago? In: International Conference on Enterprise Resource Planning Systems, pp. 42–56. Springer (2016)
- [33] Maksai, A., Bogojeska, J., Wiesmann, D.: Hierarchical incident ticket classification with minimal supervision. In: 2014 IEEE International Conference on Data Mining, pp. 923–928. IEEE (2014)

- [34] Pittke, F., Leopold, H., Mendling, J.: Spotting terminology deficiencies in process model repositories. In: *Enterprise, Business-Process and Information Systems Modeling*, pp. 292–307. Springer (2013)
- [35] Sánchez-Ferreres, J., Carmona, J., Padró, L.: Aligning textual and graphical descriptions of processes through ilp techniques. In: *International Conference on Advanced Information Systems Engineering*, pp. 413–427. Springer (2017)
- [36] Scheer, A.W., Thomas, O., Adam, O.: Process modeling using event-driven process chains. *Process-aware information systems* **119** (2005)
- [37] Selway, M., Grossmann, G., Mayer, W., Stumptner, M.: Formalising natural language specifications using a cognitive linguistic/configuration based approach. *Information Systems* **54**, 191–208 (2015)
- [38] Sinha, A., Paradkar, A.: Use cases to process specifications in Business Process Modeling Notation. In: *IEEE International Conference on Web Services*, pp. 473–480 (2010)
- [39] Smirnov, S., Weidlich, M., Mendling, J., Weske, M.: Action patterns in business process model repositories. *Computers in Industry* **63**(2), 98–111 (2012)
- [40] Weidlich, M., Dijkman, R., Mendling, J.: The icop framework: Identification of correspondences between process models. In: *International Conference on Advanced Information Systems Engineering*, pp. 483–498. Springer (2010)
- [41] Winter, K., Rinderle-Ma, S.: Detecting constraints and their relations from regulatory documents using nlp techniques. In: *OTM Confederated International Conferences" On the Move to Meaningful Internet Systems"*, pp. 261–278. Springer (2018)